Lecture 4: Ethical Problem-Solving Techniques

HUM 4441: Engineering Ethics

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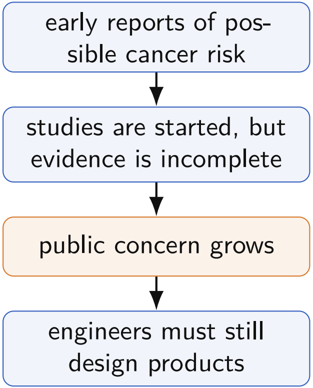
A Problem-Solving Decision Before Evidence Is Settled



In the early 1990s, concern grew that weak magnetic fields near power-distribution systems might increase cancer risk. An engineer still had to recommend whether to keep the design, reduce exposure, warn users, or wait for more evidence.

Problem-solving task

Use a method to separate the **factual issue**, the **conceptual issue**, and the **moral issue** before choosing an action.

Fleddermann, Ch. 4, p. 56

Why a Method Is Needed

Engineering habit

- define variables
- apply a model
- calculate an answer

Ethical problems

- facts may be incomplete
- concepts may be disputed
- duties may conflict

Method, not formula

Problem-solving tools do not produce automatic answers. They make the judgment clearer, more honest, and easier to defend.

Fleddermann, Ch. 4, p. 57

A Method Path

analyze **factual**, **conceptual**, **moral** issues → use **line drawing** for gray areas → use **flow charts** for consequences → handle conflicting values → apply the methods to concrete cases

These tools build in order: first sort the issue, then handle gray boundaries, consequences, and conflicts of value.

Start by Sorting the Issue

Issue type	Main question	How it is clarified
Factual	What is actually known?	evidence, measurement, records, research
Conceptual	What does this term or category mean here?	definitions, boundaries, comparisons
Moral	Which ethical principle applies?	duties, rights, honesty, safety, fairness

Order matters

Many ethical disputes look moral at first, but the real disagreement may be factual or conceptual.

Fleddermann, Ch. 4, pp. 57–58

Factual Issues: What Do We Actually Know?

- Facts are not always obvious or agreed upon.
- More research may reduce disagreement, even if it does not remove uncertainty.
- For engineers, factual uncertainty often concerns risk, performance, testing, and exposure.

Login failures

A login system occasionally locks out valid users. Before calling it "*acceptable*" or "*unfair*", first ask: how often, for whom, under what conditions, and with what harm?

Evidence check: Which missing fact would most change your judgment about shipping the login system: failure rate, affected user group, available workaround, or business deadline? Why?

Conceptual Issues: What Counts as What?

Conceptual issues are about meaning and classification.

- Is a vendor lunch a gift or a bribe?
- Is unreleased product information proprietary?
- Is a planned product an *"existing system"*?

Label	What follows
Gift	usually permissible if modest and transparent
Bribe	must be refused and reported through proper channels
Proprietary	cannot be shared casually

Why this matters

If the classification is dishonest, the moral conclusion will be dishonest too.

Fleddermann, Ch. 4, p. 58

Moral Issues: Which Principle Applies?

Once the facts and concepts are clear, the moral issue is often simpler.

- If the act is lying, honesty applies.
- If the act creates preventable danger, safety applies.
- If the act unfairly influences a decision, integrity applies.

Decision rule

Do not rush to moral judgment before checking whether the disagreement is really about facts or definitions.

From Categories to a Case

Sorting issues is useful only if it changes how a case is handled. Paradyne shows the sequence clearly: first establish what happened, then decide what the disputed action should be called, then apply the moral principle.

Why sorting comes first

The hard part is not saying *"deception is wrong"*. The hard part is deciding whether the business practice in front of you really is **deception**.

The case also raises a conflict-of-interest question: Paradyne used a former SSA employee to help lobby the agency that had issued the contract.

Paradyne Computers: Sort the Issue First

SSA request: existing systems only → Paradyne bids a system not yet running → demo uses relabeled equipment → Is this sales practice or deception?

Factual	The request required existing systems; the proposed system was not operating as proposed.
Conceptual	Does presenting a planned system as off-the-shelf count as deception?
Moral	If the action is deceptive, ordinary business pressure does not make it ethical.

Fleddermann, Ch. 4, pp. 58–59

Where Would You Press the Case?

Suppose your team is pitching software to a university. The demo works only because a hidden script bypasses unfinished modules. The client asked to see a working product.

- The team says the final product will work.
- The demo does not show the actual current system.
- The client is deciding whether to sign.

Use the three issues

What is the key factual issue?
What conceptual label is disputed?
What moral principle becomes relevant after that label is settled?

Sort Before Judging

A campus app team wants to release a bus-tracking feature before exams. During testing, the location estimate is sometimes wrong by 8–10 minutes when network coverage is poor.

The team says the feature is *"better than nothing"*. Student users may rely on it to decide when to leave for class.

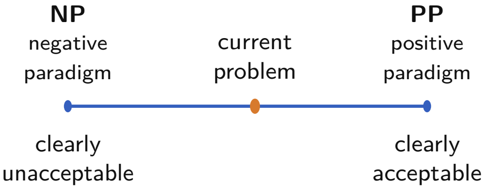
Classify the issue

Name one factual issue, one conceptual issue, and one moral issue. Which one must be clarified first?

When the Boundary Is Gray

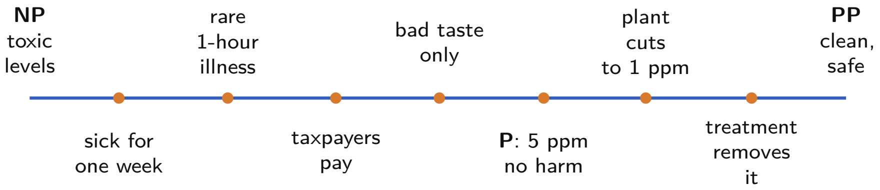
Issue sorting helps name the problem. **Line drawing** helps with the next difficulty: cases where the moral principle is clear, but the boundary is gray.

1. Set **NP** (**negative paradigm**): a clearly unacceptable case.
2. Set **PP** (**positive paradigm**): a clearly acceptable case.
3. Use obvious, fair comparison cases.
4. Place the current problem honestly.



Fleddermann, Ch. 4, pp. 59–61

Line Drawing: Waste in the Lake



Reading the line

Left means closer to the **negative paradigm**; right means closer to the **positive paradigm**. P marks the proposed 5 ppm discharge.

The proposed discharge may look permissible, but it is still far from the clearly acceptable end. Better alternatives deserve investigation.

Fleddermann, Ch. 4, pp. 59–61

Line Drawing Also Exposes Missing Facts

The line is not the end of the analysis. If placement feels uncertain, that is useful information.

In the lake case, better judgment may require:

- seasonal concentration changes
- water-use patterns in the town
- interaction with other pollutants
- likely community response

Takeaway

A good **line drawing** does not hide uncertainty. It tells you what facts would make the placement more defensible.

Fleddermann, Ch. 4, pp. 60–61

Line Drawing Can Be Abused

How it fails

- choosing biased paradigms
- placing examples dishonestly
- hiding missing facts
- treating the diagram as proof

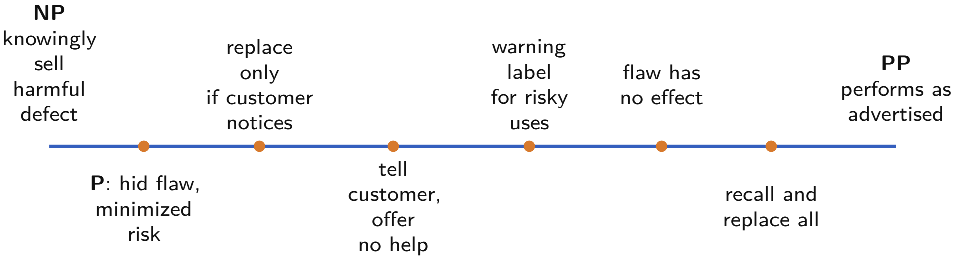
Honest use

Line drawing is useful only when the examples are chosen fairly and the uncomfortable placement is allowed to stand.

Test the placement: In the lake case, what new fact would move the proposed discharge closer to NP, the clearly unacceptable end? What fact would move it closer to PP, the clearly acceptable end?

Pentium Chip: Line Drawing

In 1994–95, Intel's Pentium chip flaw became public. The ethical issue was how Intel handled a known defect and customer information.



The book places *"tell customer, offer no help"* to the right of *"replace only if noticed"*: disclosure is better than leaving customers to discover the defect themselves.

Fleddermann, Ch. 4, pp. 61–62

Try the Same Tool on a Software Release

Your team discovers that a grade-calculation app occasionally rounds final marks incorrectly. It affects about 1 in 2,000 calculations, but only in a rare edge case. A release is scheduled tomorrow.

- Users will not notice most errors.
- The fix is likely to take three days.
- A warning could be added before release.

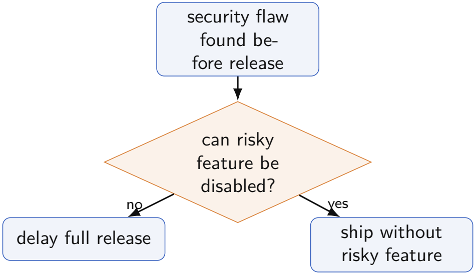
Line-drawing task

Choose the positive and negative paradigms. Then decide where "*release with a warning*" belongs. What fact would change your placement most?

When Consequences Branch

Line drawing compares cases along a boundary. **Flow charting** is different: it helps when the ethical problem depends on a sequence of decisions.

- They make consequences visible.
- They expose decision points.
- Different charts may emphasize different parts of the same case.



Fleddermann, Ch. 4, p. 62

Bhopal: Why a Flow Chart Helps

In Bhopal, methyl isocyanate was stored near a dense community. Safety systems mattered because a release could expose people outside the plant.

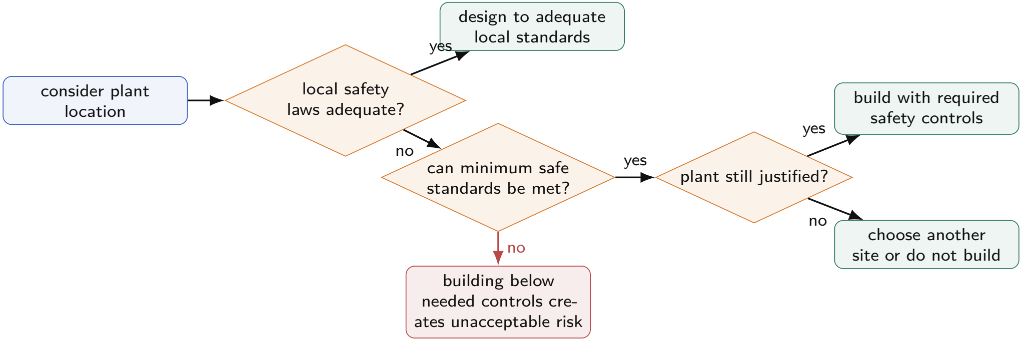
The ethical issue is not one isolated valve or one unlucky night. It is a chain of choices about location, standards, maintenance, and backup protection.

What to watch

Flow charting is useful here because each decision changes the next risk: design standard, safety equipment, maintenance timing, and emergency protection.

Fleddermann, Ch. 4, pp. 62–63

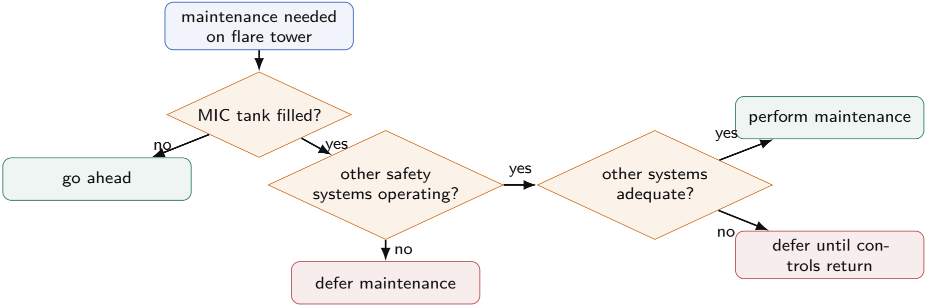
Bhopal: Location Decision Flow



This flow chart makes one point visible: **weaker local rules do not remove the engineer's duty to design for safety**.

Fleddermann, Ch. 4, pp. 62–63

Bhopal: Maintenance Decision Flow



If the tank is filled and backup systems are not adequate, the chart should end at **defer maintenance**; the method is useful only if it can block unsafe work.

Fleddermann, Ch. 4, pp. 62–63

Build a Small Flow Chart

A team finds that a password-reset feature can expose account email addresses. The full fix takes two days. The team can also disable password reset temporarily.

Build a flow chart with at least three decision/action boxes.

Required branches

What happens if the feature stays on? What happens if it is disabled? What condition would allow release?

When Two Values Pull Against Each Other

Easy priority

One value clearly outweighs the other.

Public safety normally outranks employer convenience.

Creative middle

Redesign the situation so fewer values are sacrificed.

Reduce waste, pretreat it, or fund treatment.

Hard choice

If no middle path works, make the best defensible choice with the information available.

Fleddermann, Ch. 4, pp. 63–64

Challenger: A Conflict With Lives at Stake

Thiokol manager Bob Lund faced pressure before launch. Engineers worried that cold weather could prevent rubber O-rings in the booster joints from sealing.

Reasons not to launch

- cold-weather sealing concern
- possible loss of crew
- uncertainty in the data

Pressure to launch

- schedule pressure
- NASA relationship
- jobs and future contracts

The conflict

Jobs, contracts, and schedule cannot be balanced against preventable risk to human life as if they were equal units.

Fleddermann, Ch. 4, p. 64

What Would Count as a Middle Way?

In the Challenger case, one possible middle path would have been delay. Another would have been making the risk explicit to those asked to bear it.

But a middle way is not automatically acceptable. It must actually reduce the ethical problem, not just make the decision easier to defend afterward.

Security-release choice

If a software team finds a security flaw before launch, which is the stronger middle way: delay release, release with a warning, disable the risky feature, or ask users to accept the risk? What fact decides?

Gifts, Favors, and Bribery

Gifts, favors, and bribes sit on a gray professional boundary where line drawing is useful.

A bribe is a benefit offered to someone in a position of trust to influence dishonest action.

Bribery is wrong because it:

- distorts fair competition
- favors those with money and access
- turns professional judgment into something for sale

Main boundary

A **gift** becomes ethically dangerous when its value, timing, or intent can influence judgment or create the appearance of influence.

Bribery is illegal even when local business culture tolerates it. U.S. companies are also barred from bribery overseas.

Fleddermann, Ch. 4, pp. 65–66

Place the Gift on the Line

Scenario	Likely issue	What to ask
Logo coffee mug from vendor	low-value advertising	Is it nominal and transparent?
Expensive crystal bowl	value and intent	Would it influence future purchase decisions?
Vendor pays for expensive lunch	unequal benefit	Is business judgment being bought?
Supplier pays for conference trip with family invited	personal benefit	Could this be defended publicly?

Place one case: Pick one scenario. Place it on a line from *"nominal gift"* to *"bribe"*. What fact would move it: value, timing, secrecy, decision authority, or repeated pattern?

Fleddermann, Ch. 4, pp. 65–67

Avoiding Bribery Problems

Policy

Follow company rules on gifts, meals, travel, and vendor contact.

Approval

When rules are unclear, ask before accepting the benefit.

Public test

If you would struggle to defend it publicly, do not do it.

Appearance matters

Even if no decision is actually sold, the appearance of impropriety can damage trust and distort future choices.

Fleddermann, Ch. 4, p. 67

Spiro Agnew: Kickbacks as a System

In Maryland public-works contracting, engineering firms paid kickbacks to secure government contracts. Lester Matz's firm paid Spiro Agnew a percentage of fees from county and state work; the payments continued as Agnew rose politically.

The ethical issue is not only one payment. A kickback system changes what gets rewarded: access replaces competence.

Analyze it

Factual: what payments were made? **Conceptual**: campaign support, gift, or kickback? **Moral**: what happens to fair competition and public trust?

Fleddermann, Ch. 4, pp. 69–70

Cell Phones and Cancer: Managing the Unknown

Design decisions often happen before science has settled every risk question. The cell-phone case is not a disaster after the fact. It is a design problem under scientific uncertainty.

- studies did not show clear harm
- measurement and long-term effects remained difficult
- redesign could reduce emissions but cost more

Engineering duty

Be informed about potential risks and reduce them where reasonable, even before every scientific question is settled.

Fleddermann, Ch. 4, pp. 67–69

Job Searches and Assignments Count Too

Job search

If a company pays for a visit, honesty matters: are you seriously considering the company, or using the trip for personal benefit?

Assignments

Cheating is not only a school rule issue. It trains dishonesty into professional work.

Classify the issue: A teammate asks to copy your finished code because the deadline is close. What are the factual issue, conceptual issue, and moral issue?

Fleddermann, Ch. 4, pp. 70–71

Lecture 4 Summary

- Ethical problem solving is disciplined judgment, not formula calculation.
- Start by separating **factual**, **conceptual**, and **moral** issues.
- Use **line drawing** when the principle is clear but the boundary is gray.
- Use **flow charting** when consequences depend on a chain of decisions.
- Conflict problems require priority, creative middle paths, or a hard defensible choice.
- Gifts and bribery show why value, timing, intent, policy, and public appearance all matter.
- In CS work, the same tools apply to bugs, demos, security flaws, data systems, teamwork, and academic honesty.

References

- Charles B. Fleddermann, *Engineering Ethics*, 4th ed., Chapter 4.